```
  end
end
```

Run the following command in your Terminal to apply the Migration.

```
bin/rails db:migrate
```

After adding the index, open psql for Rideshare and run `\d users` to confirm that you now see a new GIN index covering the `users.first_name` field.

After adding the index and a bit of warm-up, the same ILIKE query from above should be much faster! Confirm the index is used by adding `EXPLAIN ANALYZE` in front of your query in psql.

Next you'll see another new concept. PostgreSQL helps you explore the similarity and distance between words. Run the query below to see how similar "henr" is to each of the names. The distances between "henr" and the name can also be printed. The % operator[10] is being used in the `WHERE` clause to filter the results.

sql/query_similarity_distance.sql

```
SELECT
    distinct first_name,
    similarity(first_name, 'henr') AS similarity,
    first_name <-> 'henr' AS distance
FROM users
WHERE first_name % 'henr';

--  first_name | similarity | distance
-- ------------+------------+-----------
--  Henrietta  | 0.36363637 | 0.6363636
--  Henriette  | 0.36363637 | 0.6363636
--  Henry      |  0.5714286 | 0.4285714
```

If you hadn't guessed it, the query above is also using the new GIN index you created.

Next you'll change things up a bit.

In the previous query a fragment of a word was matched exactly. What about similar words that don't match exactly? For example, a search query from a user may have a typo. For typo errors where the letters are ordered incorrectly, the search would produce no results which is not a good user experience. We all make typos!

As an example, searching "Henrieta" with one t instead of two will not produce any results.

10. https://www.postgresql.org/docs/9.1/pgtrgm.html